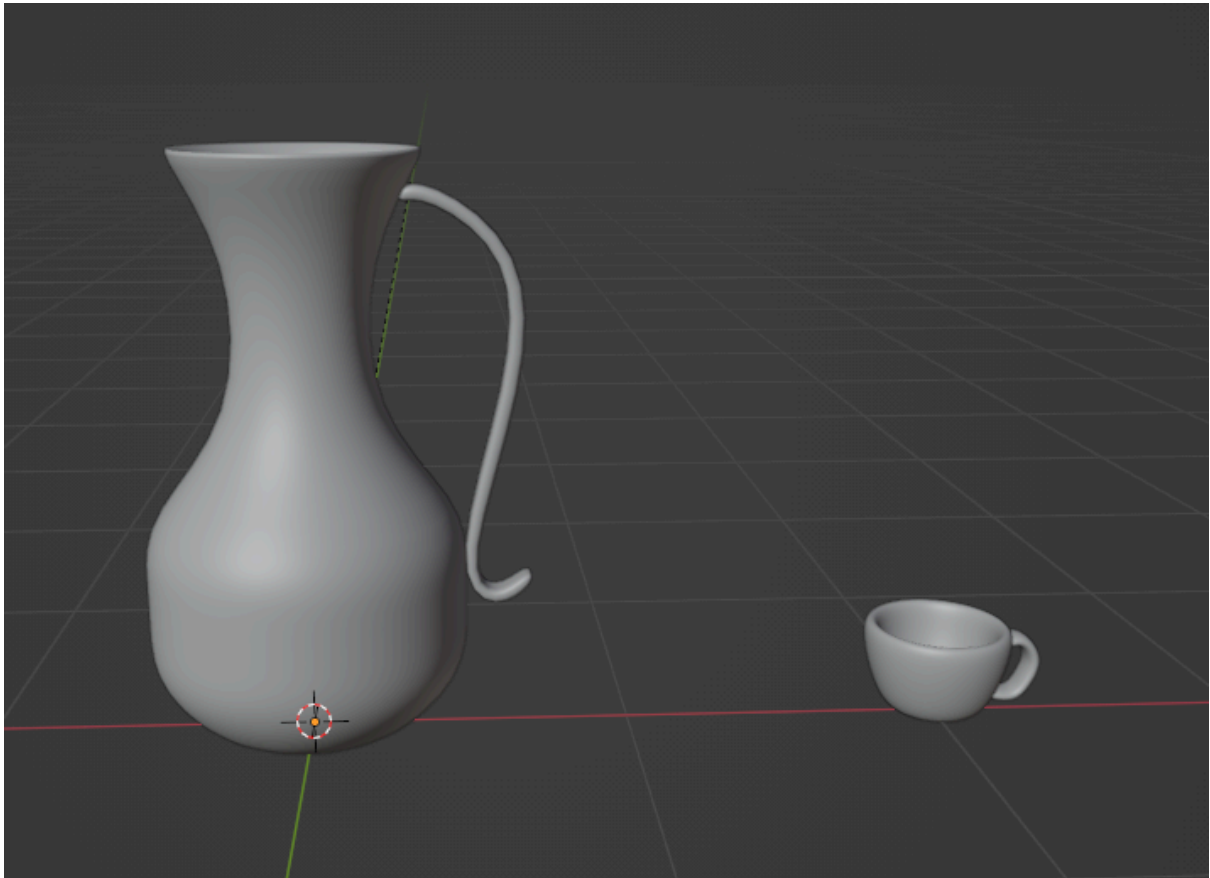
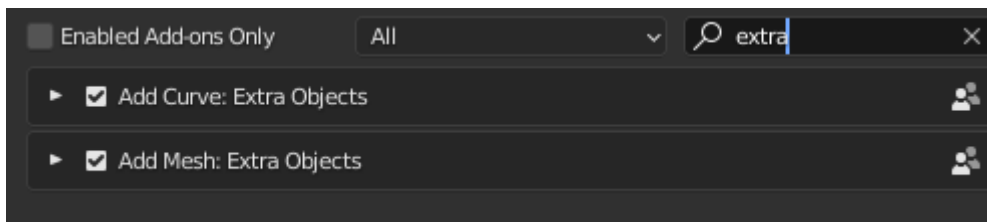


2 Simple Models Tutorial

Important concepts covered: screw modifier, extrude vertices, solidify modifier

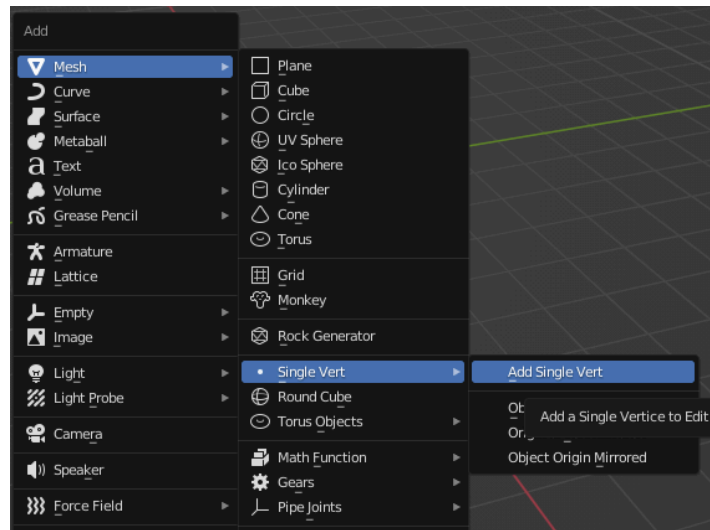


First, go to Edit > Preferences > Add-ons and search for Extra Objects and check both “Add Curve and Mesh:

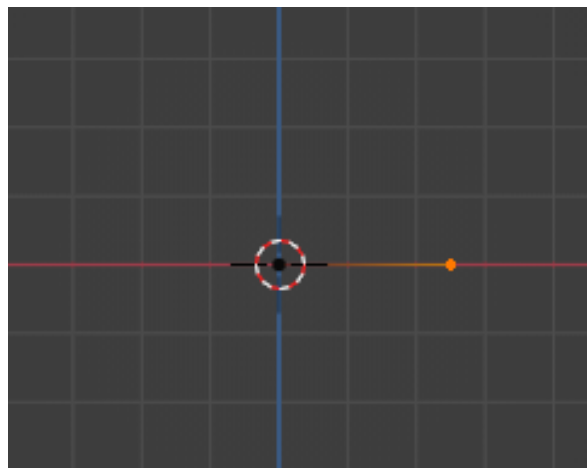


Before starting the system, make sure to save the project first!

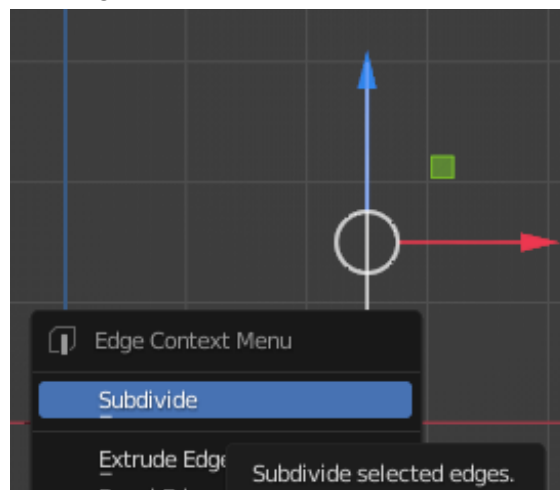
Select all the objects of the scene (press “A” to select all) and delete (DEL or X). Then, add a “Single vert” mesh (SHIFT+A → MESH → Single Vert → Add Single Vert):



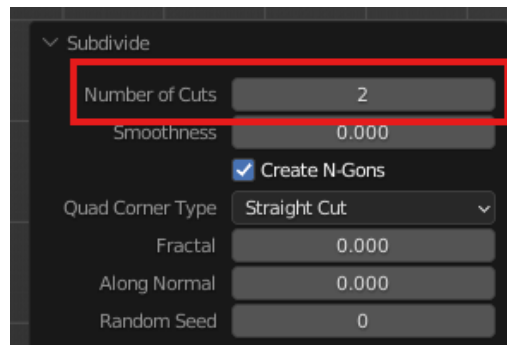
Blender should automatically enter in Edit mode. Go to the front view by pressing “NUMPAD 1”. With the vertex selected, extrude (E) it and move in the X direction (G+X) by 0.25



Now extrude (E) this vertex and move it along the Z axis (G+Z) by 0.3. Then, change to edge selection mode and select the vertical edge created and subdivide it by clicking with the right mouse button and then selecting “Subdivide”:



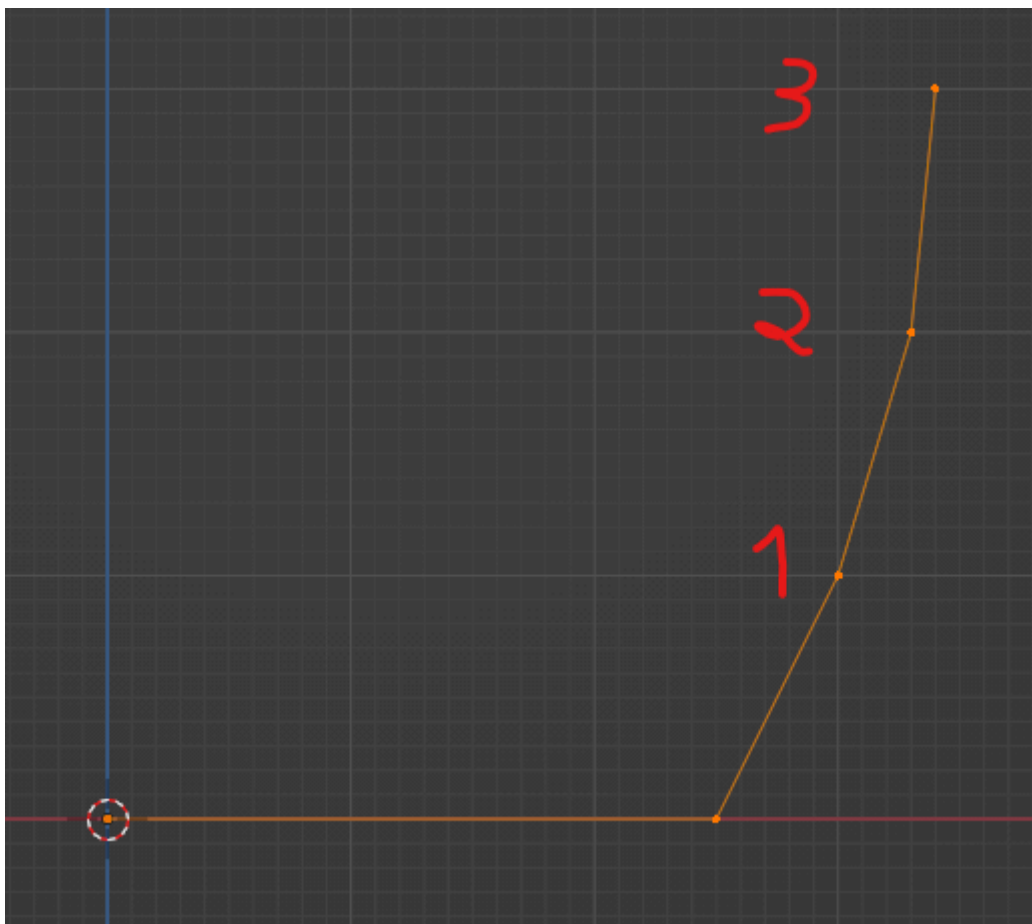
Then, on the context menu that appears on the bottom left corner of the screen, set “Number of Cuts” to 2:



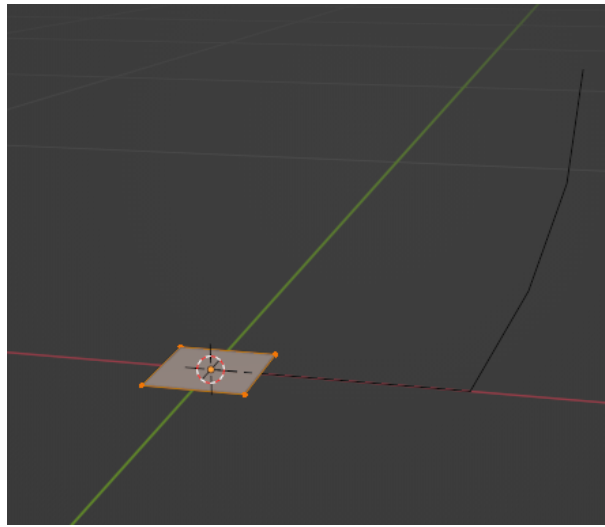
Change to vertex selection mode and select:

- The first vertex created (from the bottom up) and move it along the x axis (G+X) by 0.05,
- The second vertex created (from the bottom up) and move it along the x axis (G+X) by 0.08,
- The third vertex created (from the bottom up) and move it along the x axis (G+X) by 0.09.

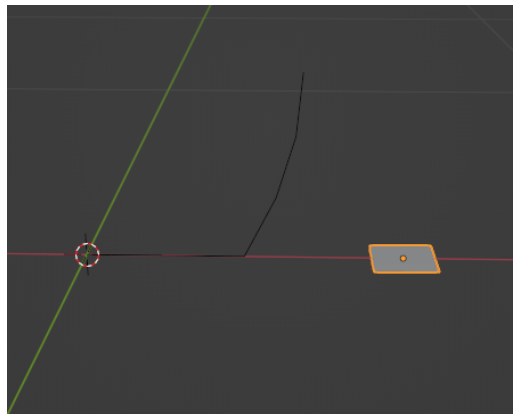
You should have something like this:



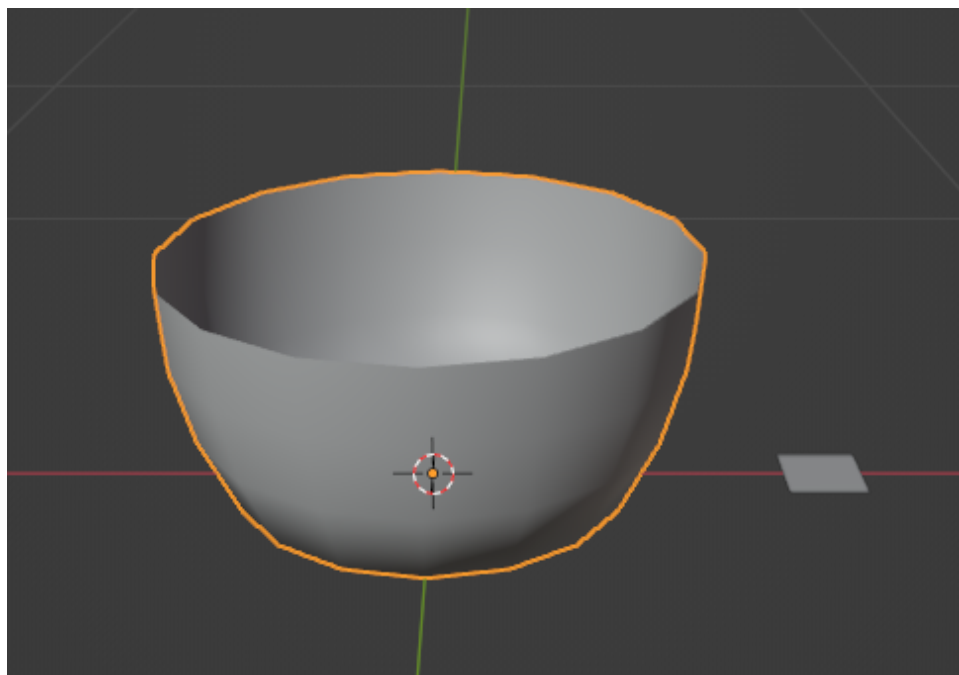
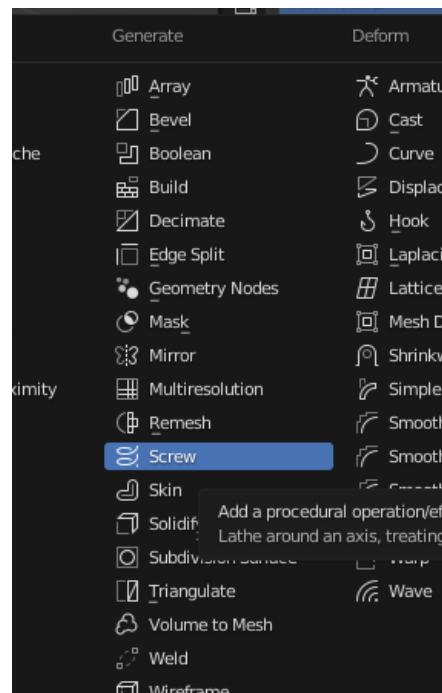
Now go back to Object mode ("TAB"), make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a new Plane mesh (SHIFT+A → MESH → Plane). enter edit mode on the plane created and scale it (S) down by 0.05:



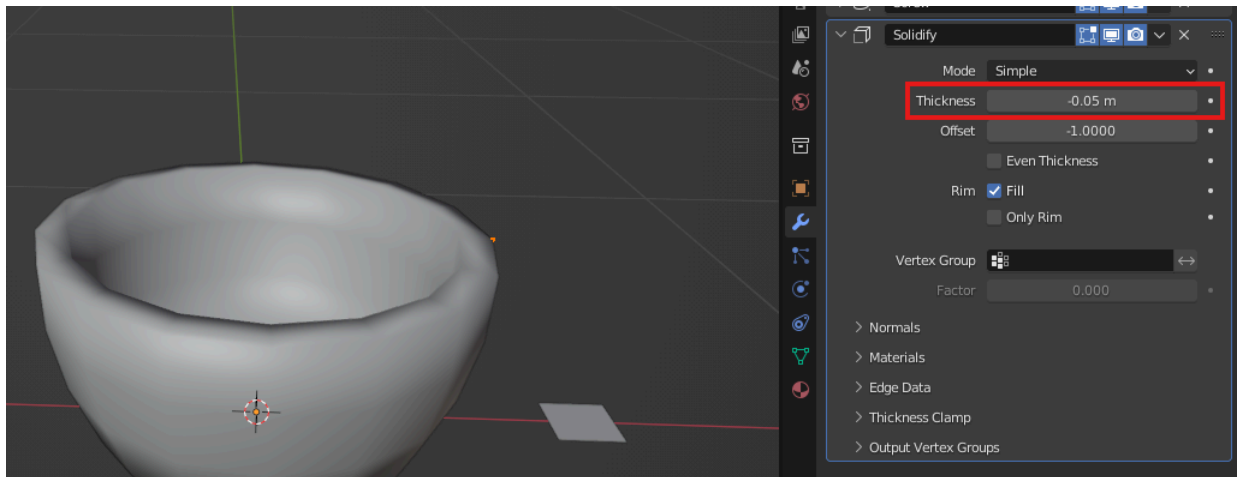
Go back to object mode again and move the plane by 0.5 along the X axis (G+X):



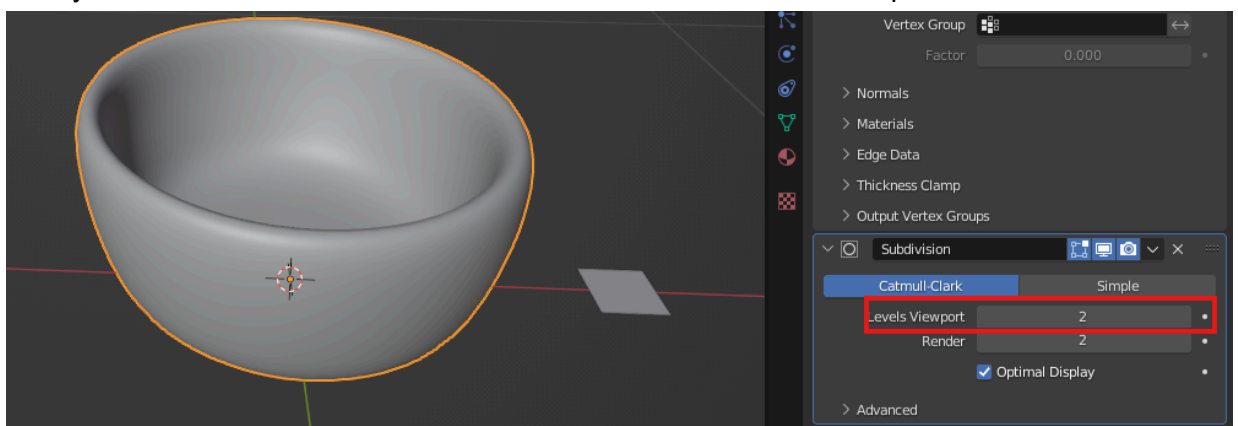
Select the other object created before and add a new Screw modifier to make the object round:



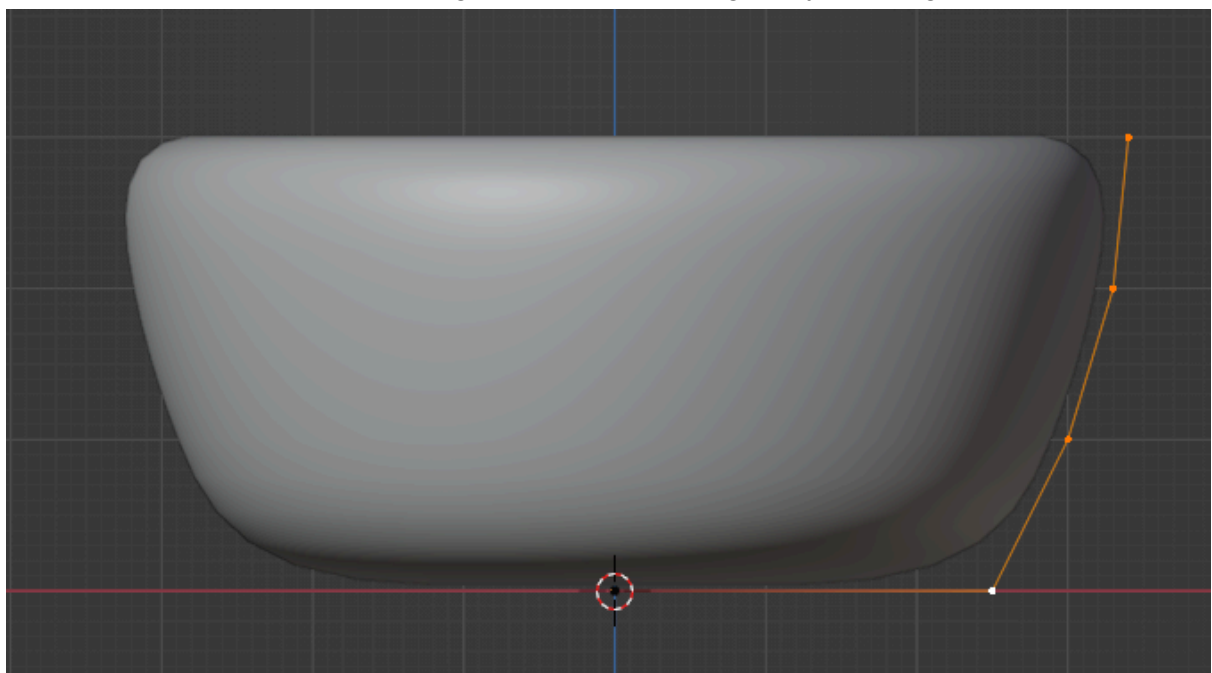
Next, add a Solidify modifier to give it some thickness, setting the “Thickness” to -0.05:



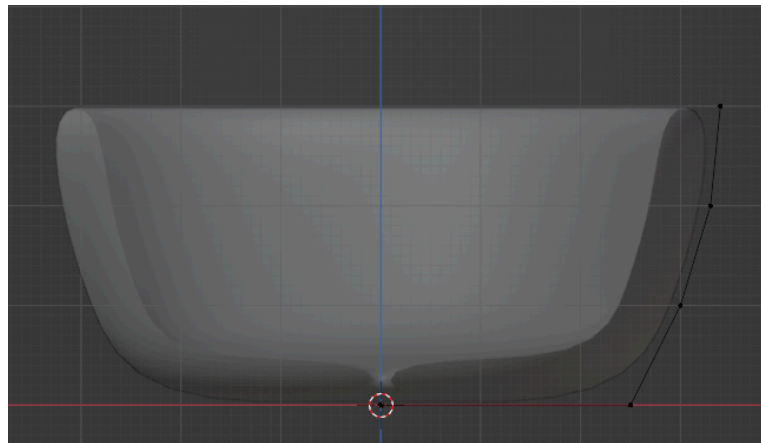
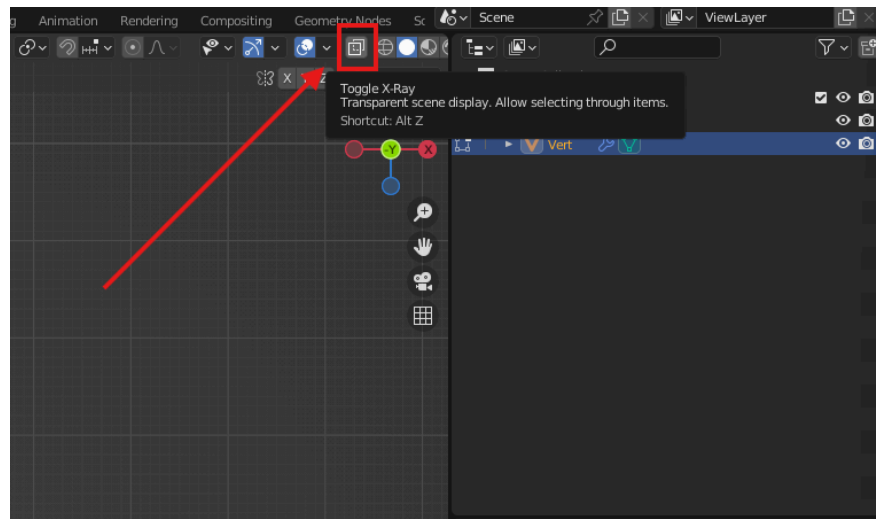
Finally, add a “Subdivision Surface” modifier and set the Levels Viewport to 2:



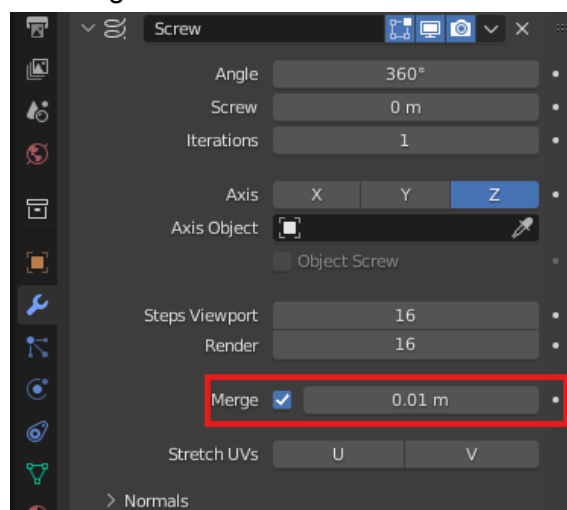
Now select it, enter edit mode and go to the front view again by pressing “NUMPAD 1”:



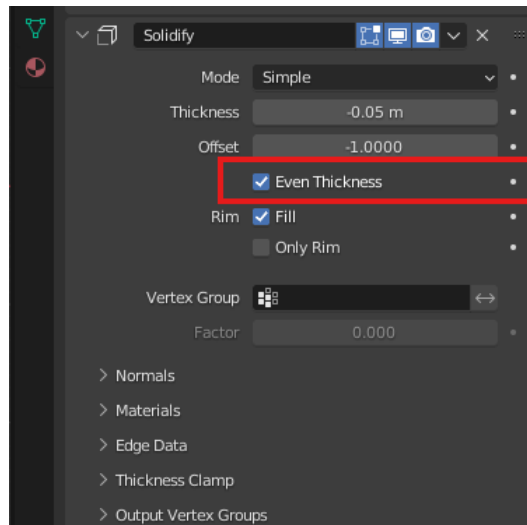
In order to see through the object, toggle x-ray mode by pressing the button on the top right corner of the screen:



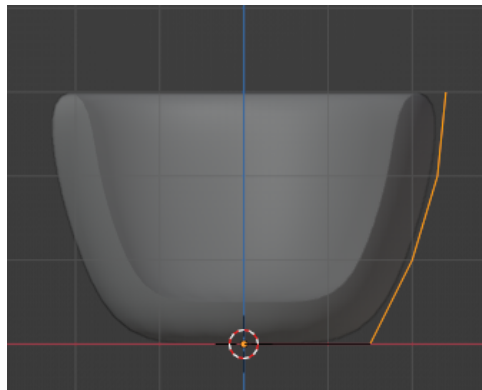
As you can see, there is a problem with the mesh: the bottom part has a hole. This is due to the screw modifier that does not automatically merge the vertices. So to fix it, go to the Screw modifier and enable “Merge”:



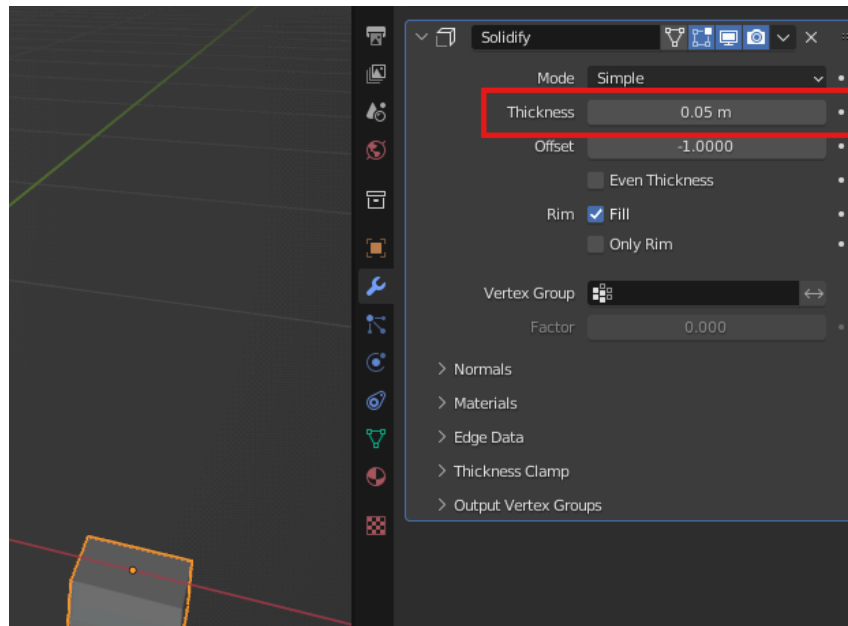
Another option we can enable is located at the Solidify modifier and is the option called “Even Thickness”. This will ensure that the thickness will be always the same to the opposite side:



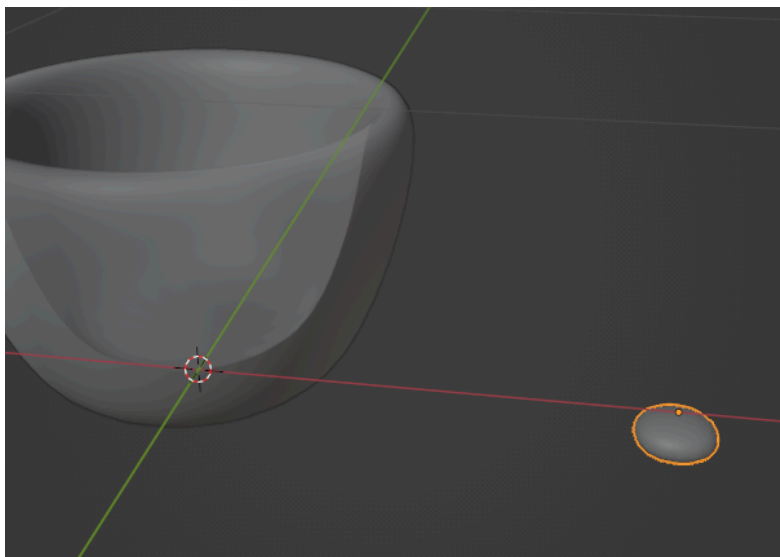
Now to make the cup better looking, change to Edge selection mode and select the vertical edges and move it along the X axis (G+X) by -0.1:



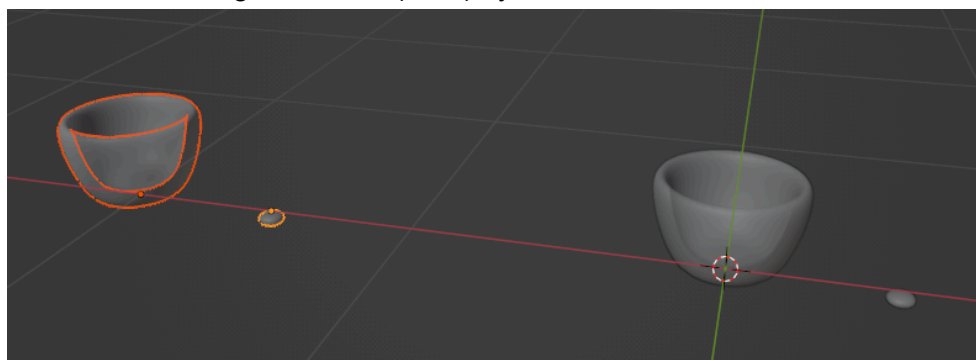
Go back to object mode and select the created plane. Add a Solidify modifier and set its Thickness to 0.05:



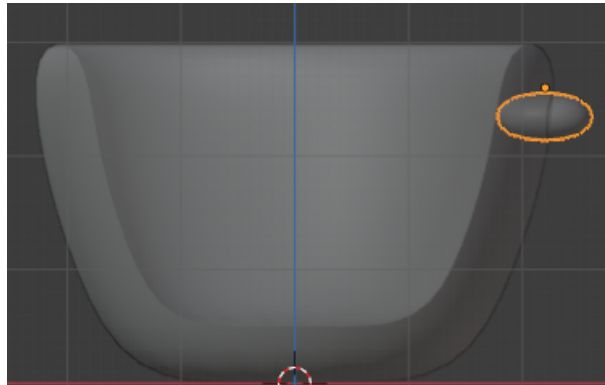
Then, as before, add a Subdivision Surface Modifier and set the Levels Viewport to 2. To make it look smoother, right click on viewport with the cube selected and select Shade Smooth.



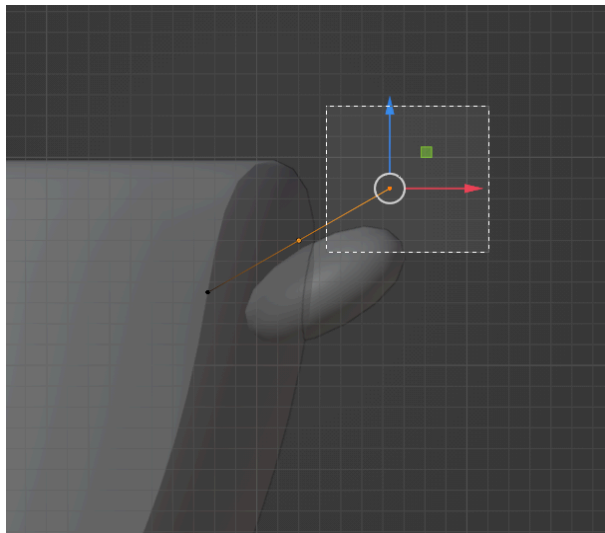
Now, select both objects (the cube and the cup, in this order) and press "SHIFT+D" to copy them and move them along the X axis (G+X) by -2:



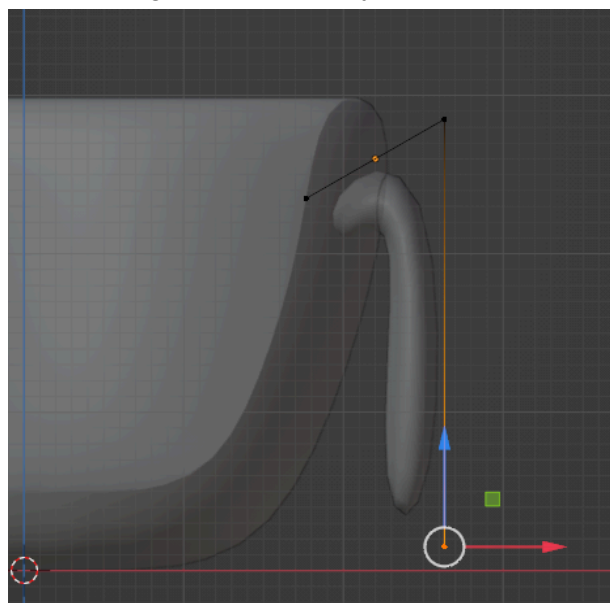
For now, we are going to ignore these copied objects, so go back to the original 2 and go to Front view ("NUMPAD 1"). Select the cube and move it along the Z axis (G+Z) by 0.26 and along the x axis (G+X) by -0.28:



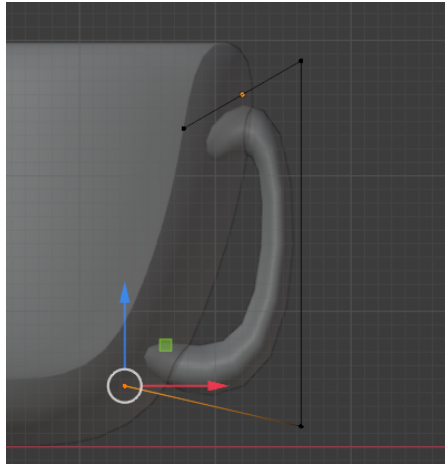
This will be the handle of the cup. Now rotate it (R) by -30 degrees and enter edit mode. Make sure you are still in x-ray mode, then change to vertex selection mode and click and drag to select the right most vertices of the handle:



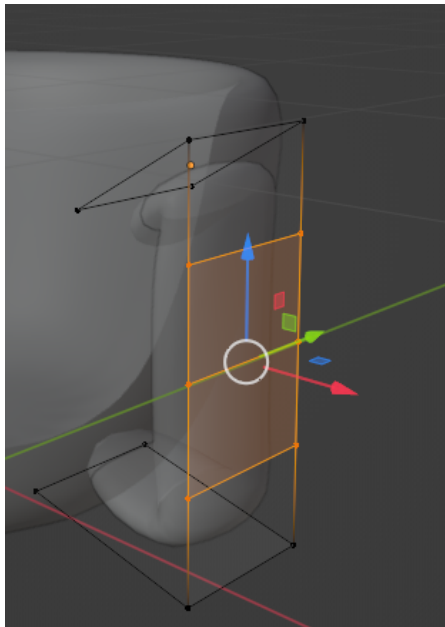
Now extrude (E) it and move along Z axis (G+Z) by -0.27:



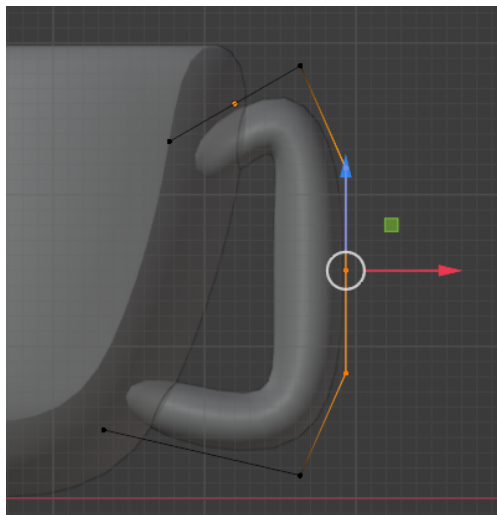
Then extrude again and move on the X axis (G+X) -0.13 and Z axis (G+Z) 0.03:



Now, go to perspective view and add 3 Loop cuts on the vertical face (you can do this by pressing “CTRL+R” and use the scroll of the mouse to increase the number of cuts and then double click to confirm):

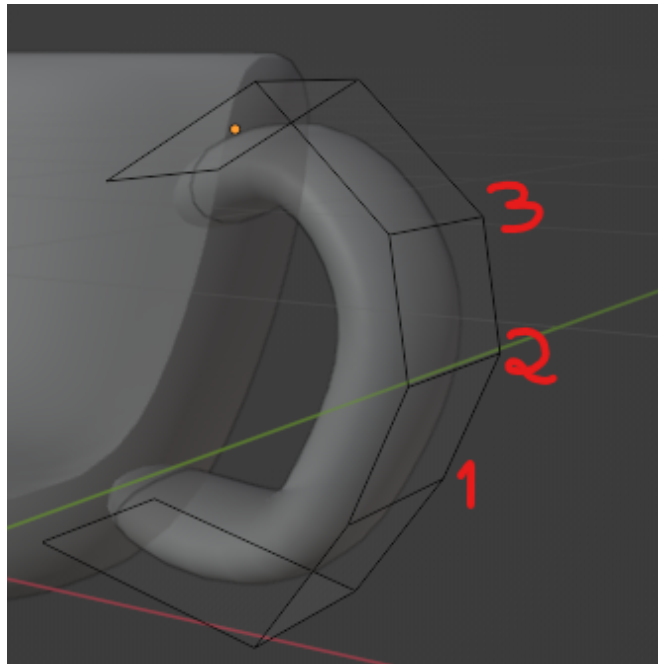


Now with this same selection shown in the image, move along the X axis (G+X) by 0.03:

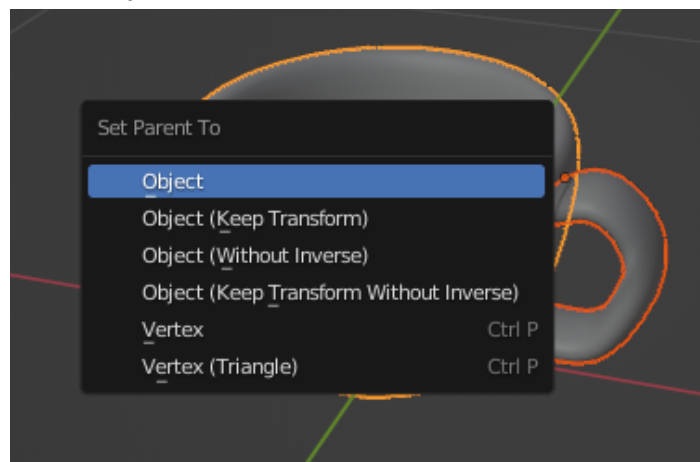


Now, change to edge selection mode and, counting from the bottom up again:

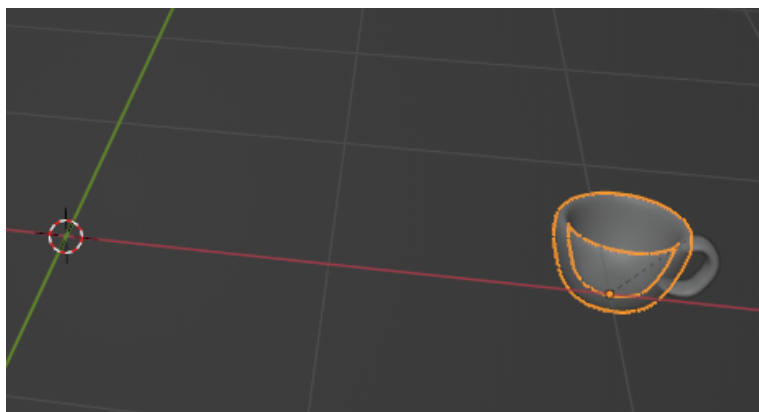
- Select the first edge and move it along X axis (G+X) by 0.02,
- Select the second edge and move it along the X axis by (G+X) 0.05,
- Select the third edge and move it along the X axis (G+X) by 0.04



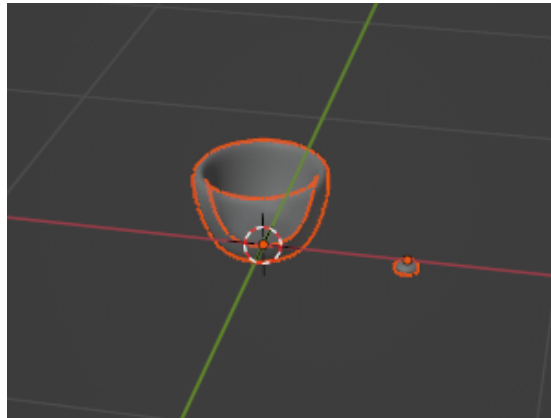
Now go back to object mode and select first the handle and then the cup (in this order) and hit “CTRL+P” and select “Object”:



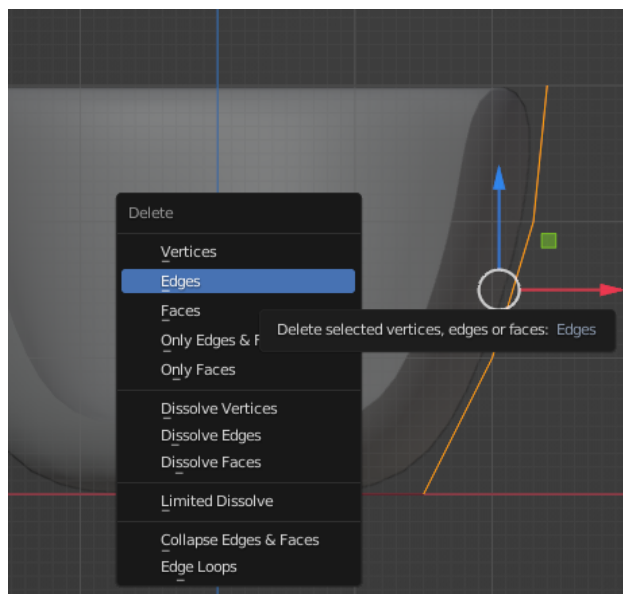
Now the handle will follow the cup. So select the cup and move it along X axis (G+X) by 2:



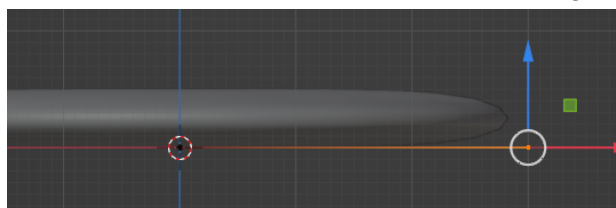
Now we are going to create a big jar. To start, select the 2 objects (the cube and then the jar, in this order) that we've duplicated and decided to ignore before and move in the X axis (G+X) by 2:



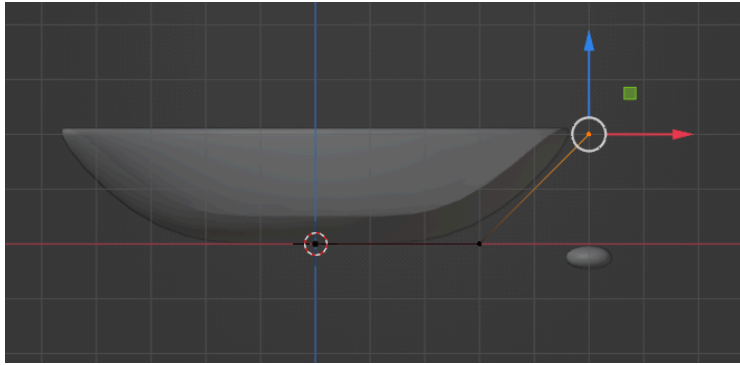
Now select the jar, go to the front view again and enter edit mode. Delete (DEL or X) all the vertical edges:



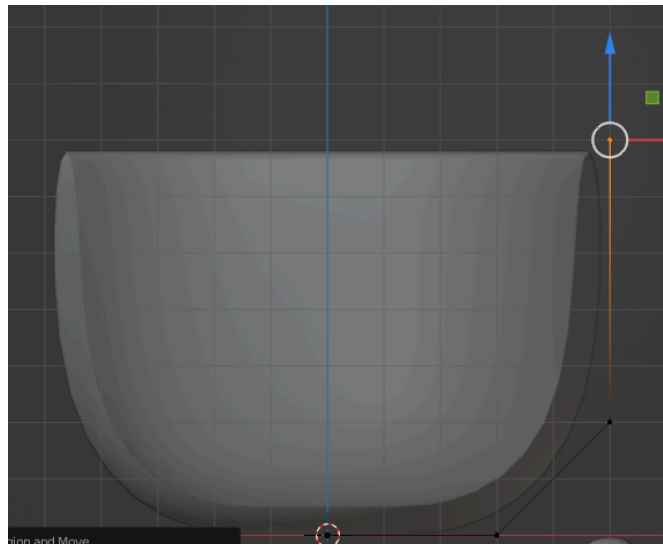
Go to vertex select mode, select the outer vertex and move it along X axis (G+X) by 0.15:



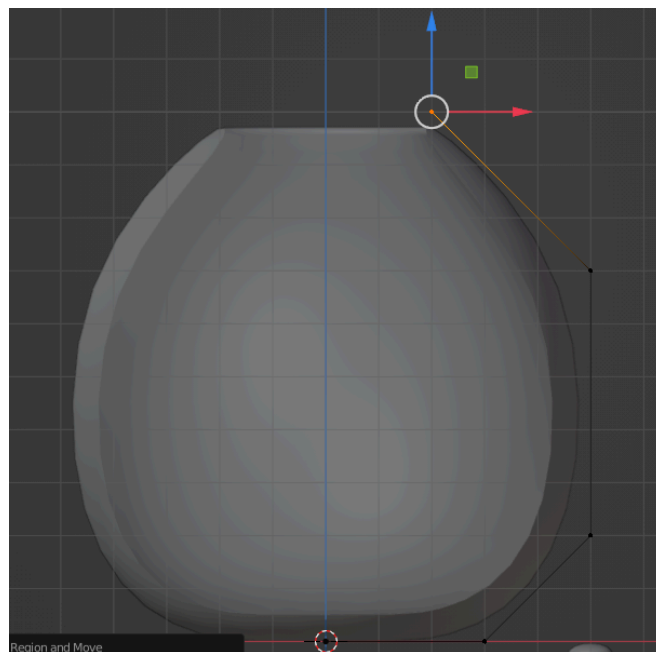
Extrude (E) it and move X = 0.2 (G+X) and Z = 0.2 (G+Z):



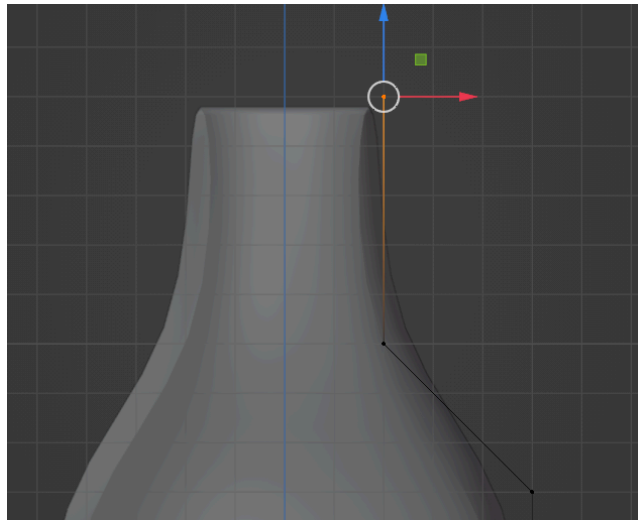
Now extrude (E) again and move along the Z axis (G+Z) by 0.5:



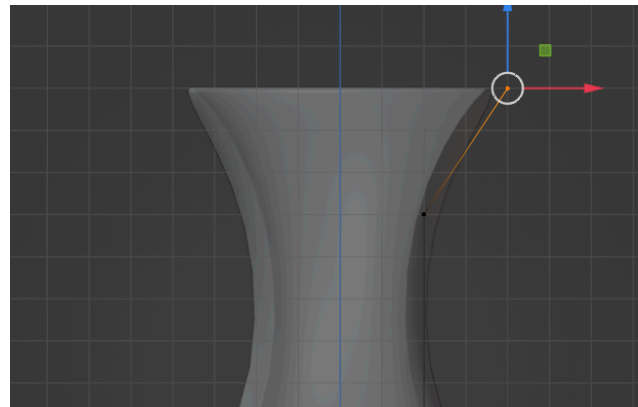
Next, extrude (E) again and move X = -0.3 (G+X) and Z = 0.3 (G+Z):



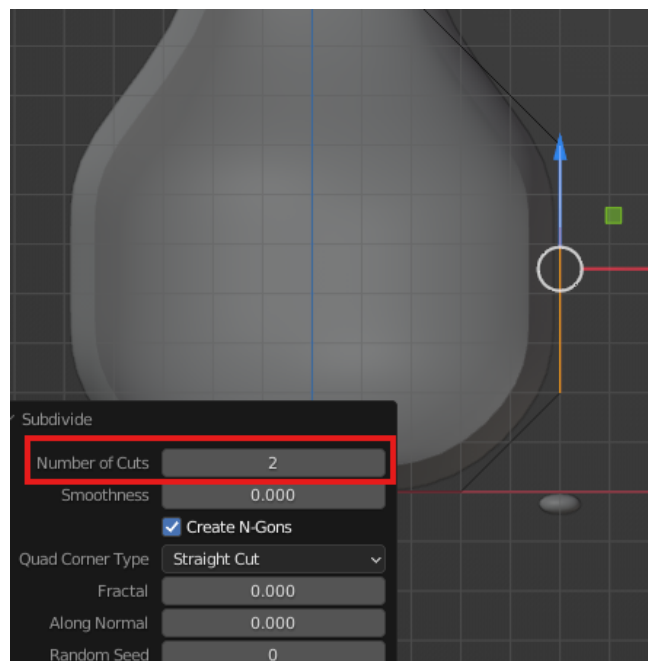
Extrude (E) again and move along Z axis (G+Z) by 0.5:



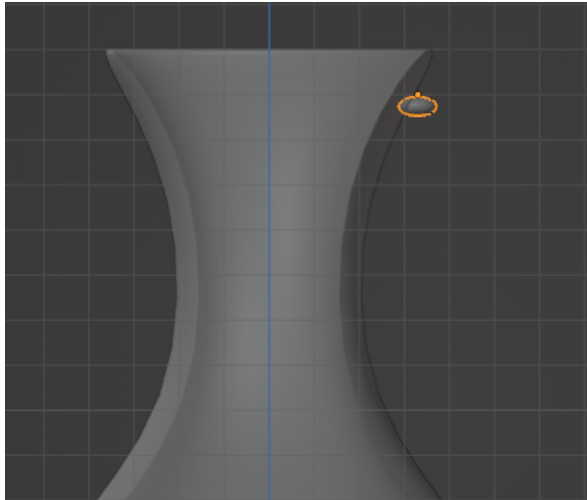
Extrude (E) again and move $X = 0.2$ (G+X) and $Z = 0.3$ (G+Z):



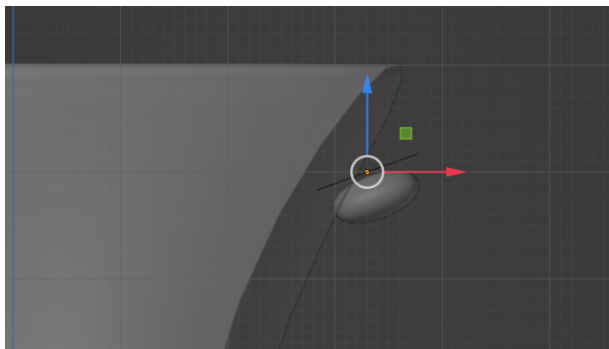
Now, to make the bottom part more “squared”, change to edge selection mode and select the outermost vertical edge, Right click and select “Subdivide” and then set the Number of Cuts to 2:



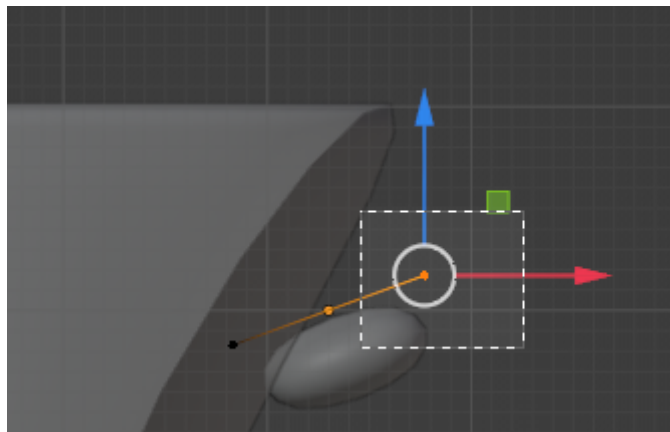
Now go back to object mode and select the square object (that will be the handle again) and move it $Z = 1.7$ (G+Z) and $X = -0.17$ (G+X):



Rotate (R) it by -20 and enter edit mode:



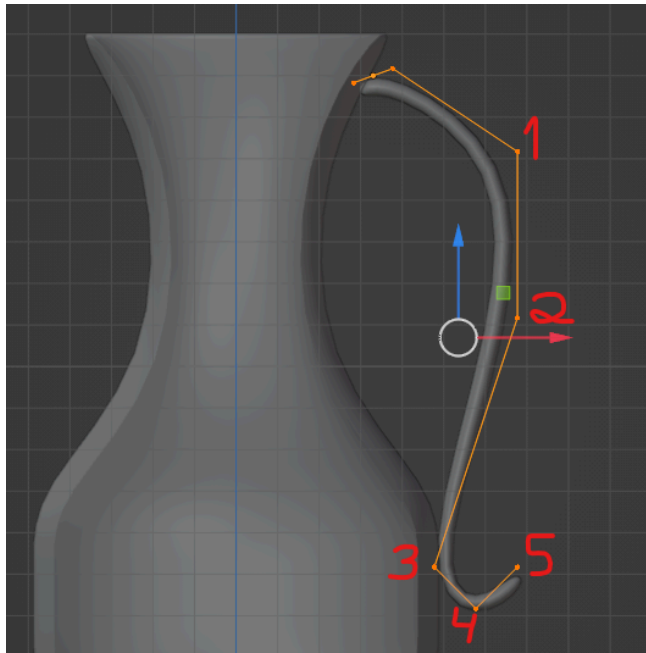
Make sure you have X-ray enabled and you are in the front view. Change to vertex mode, click and drag to select the outermost vertices:



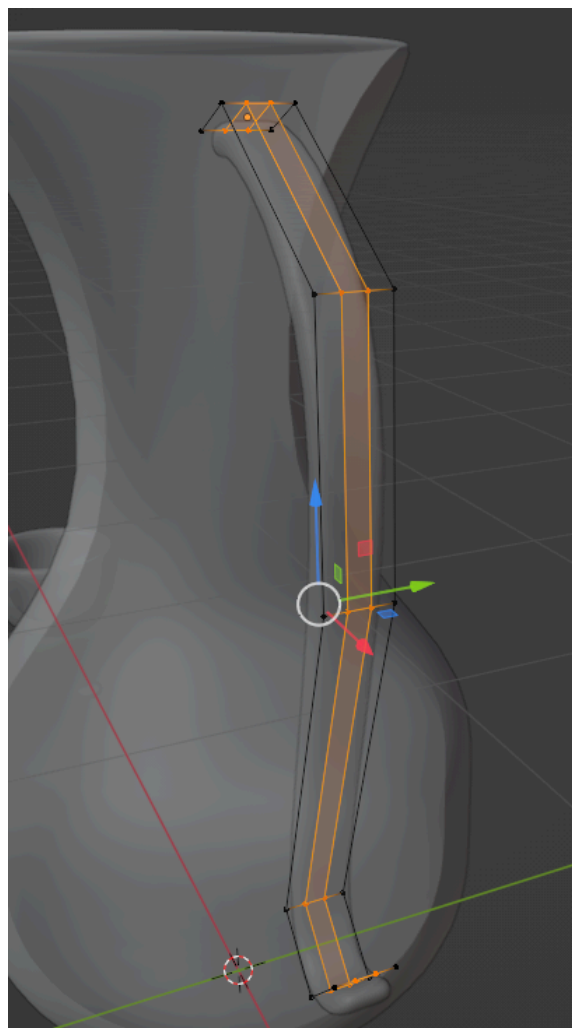
With this selection, repeat the following steps to create the long handle:

- (1) Extrude (E) and move $X = 0.3 (G+X)$, $Z = -0.2 (G+Z)$,
- (2) Extrude (E) and move $Z = -0.4 (G+Z)$,
- (3) Extrude (E) and move $X = -0.2 (G+X)$, $Z = -0.6 (G+Z)$,
- (4) Extrude (E) and move $X = 0.1 (G+X)$, $Z = -0.1 (G+Z)$,
- (5) Extrude (E) and move $X = 0.1 (G+X)$, $Z = 0.1 (G+Z)$

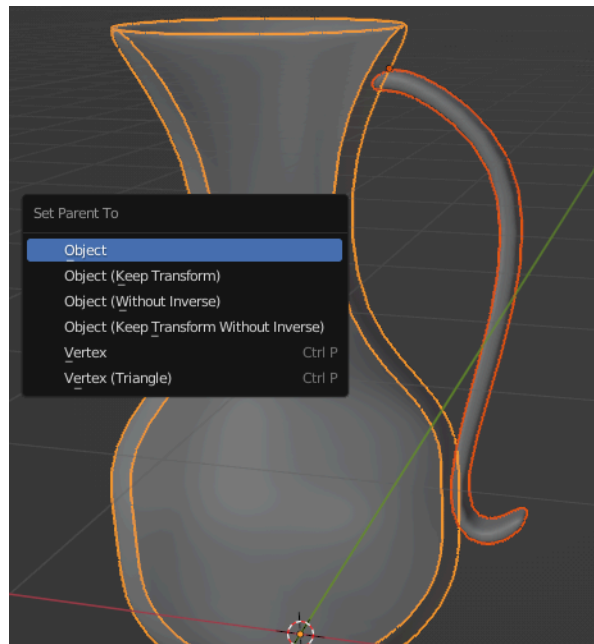
You should end up with the following mesh:



Next, go to perspective mode and add a Loop Cut (CTRL+R) with 2 cuts along the vertical faces of the handle:



Finally, go back to object mode, select first the handle and then the jar (in this order) and, like before, press “CTRL+P” and select “Object” to parent them:



Now you can disable x-ray mode.

